Machine cycle times: from linear to hierarchical models

Let Y_{ij} be the cycle time (seconds) of production run i on machine j ($j = 1, \ldots, 24$), X_{ij} the hours since the machine's last maintenance, and $N_{ij} = 1$ if the run was on the night shift (0 for day). Machines are spread over three plants and have between 3 and 20 runs each.

Synthetic data

The data were simulated as

$$Y_{ij} = 60 + b_{0j} + (0.05 + b_{1j})\, X_{ij} + 1.5\, N_{ij} + \varepsilon_{ij}$$
$$b_{0j} \sim N(0,\ 4^2), \qquad b_{1j} \sim N(0,\ 0.02^2), \qquad \varepsilon_{ij} \sim N(0,\ 2^2)$$

So every machine has its own baseline and wear rate, the night penalty is the same 1.5 s for all machines, and there is no true plant effect.

M1: Simple linear model (complete pooling)

$$Y_{ij} \mid \beta_0, \beta_1, \sigma \sim N(\beta_0 + \beta_1 X_{ij},\ \sigma^2)$$

One line for every machine; only tool wear is used.

M2: Multiple linear model (still pooled)

$$Y_{ij} \mid \beta, \sigma \sim N(\beta_0 + \beta_1 X_{ij} + \beta_2 N_{ij} + \beta_3\, \mathrm{PlantB}_j + \beta_4\, \mathrm{PlantC}_j,\ \sigma^2)$$

Adds shift and plant, but machines within a plant are still treated as identical.

M3: Categorical machine (no pooling of intercepts)

$$Y_{ij} \mid \beta_{0j}, \beta_1, \beta_2, \sigma \sim N(\beta_{0j} + \beta_1 X_{ij} + \beta_2 N_{ij},\ \sigma^2)$$
$$\beta_{0j} \sim N(\cdot, \cdot) \text{ independently for each } j$$

Each machine gets its own baseline (23 dummy variables); machines share nothing.

M4: Hierarchical, varying intercepts (partial pooling)

$$Y_{ij} \mid \beta_{0j}, \beta_1, \beta_2, \sigma \sim N(\beta_{0j} + \beta_1 X_{ij} + \beta_2 N_{ij},\ \sigma^2) \quad \text{(within machine)}$$
$$\beta_{0j} \mid \beta_0, \sigma_0 \sim N(\beta_0,\ \sigma_0^2) \quad \text{(between machines)}$$
$$\beta_0,\ \beta_1,\ \beta_2 \sim N(\cdot, \cdot), \quad \sigma \sim \mathrm{Exp}(\cdot), \quad \sigma_0 \sim \mathrm{Exp}(1)$$

Machine baselines come from a common population, so each is pulled (shrunk) toward β_0; machines with few runs are pulled the most.

M5: Hierarchical, varying shift effect

$$Y_{ij} \mid \beta_{0j}, \beta_1, \beta_{2j}, \sigma \sim N(\beta_{0j} + \beta_1 X_{ij} + \beta_{2j} N_{ij},\ \sigma^2)$$
$$\begin{pmatrix} \beta_{0j} \\ \beta_{2j} \end{pmatrix} \sim N\left(\begin{pmatrix} \beta_0 \\ \beta_2 \end{pmatrix},\ \Sigma \right)$$

Each machine also gets its own night-shift penalty. In the R code `(shift | machine)` has no fixed `shift` term, so there $\beta_2 = 0$ and the night penalties are centred on zero.

Model comparison (leave-one-out ELPD; higher is better)

Model	ELPD diff	SE diff
M4 Hierarchical, varying intercepts	0.0	0.0
M3 Categorical machine	−0.9	1.4
Linear: hours + shift	−130.3	12.0
M4 Hierarchical, varying intercepts	0.0	0.0
M5 Hierarchical, varying shift	−2.7	4.9

Reading the results

- The pooled linear model ignores the 4 s spread between machines and loses decisively (−130.3, about 10 standard errors).
- M3 and M4 both capture machine differences and predict about equally well. M4 does it with one variance parameter, σ_0, instead of 23 dummies, and can predict a *new* machine; M3 cannot.
- M5 adds machine-specific night penalties that do not exist in the data, so the extra parameters fit noise. Shrinkage keeps the cost small (−2.7, within one standard error), and the simpler M4 is preferred.